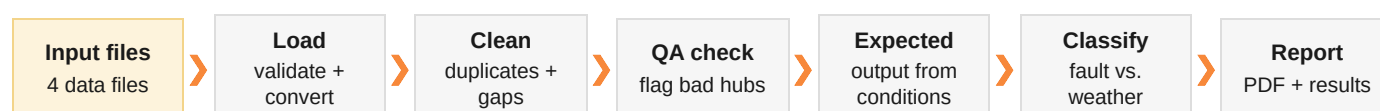




From panel output and conditions to every output drop, **explained**

1 THE PROBLEM

When a commercial array's output drops, O&M teams can't tell a failing panel from a snowy week, so they roll a truck to find out. SolarScope hubs measure two things on every panel: **output** (voltage and current) and **conditions** (the panel's microclimate). This team builds SolarScope's first MVP, the pipeline that turns those two streams into an explanation for every drop.



2 CRITERIA

- **One command.** The pipeline runs on a laptop from input folder to finished PDF.
- **Handles messy data.** Duplicates, gaps, and out of range values are flagged, never deleted.
- **Catches the failing sensor.** A hub whose sensor drifts after the January snowstorm is excluded automatically, so a bad sensor is never called a bad panel.
- **Fault or weather.** Every drop gets a label. Snow, soiling, cloud, and heat are weather. Hotspots, string outages, and wiring issues are faults. Read the voltage and current pattern, not just power.
- **Dispatch only real faults.** The dataset hides 6 seeded faults. Catch at least 5 with no more than 2 false dispatches.
- **Catch decline early.** A panel slowly losing output is flagged before it falls 10% behind its neighbors.
- **Traceable.** Every number in the report comes from a saved results file.
- **Maintainable.** Thresholds sit in a config file, every step has tests, and real data works with no code changes.

4 DATA WE PROVIDE

- **readings.csv:** Readings every 15 minutes from 12 hubs covering 36 panels. Output: voltage and current per panel. Conditions: back of panel temperature, humidity, and irradiance. About 420,000 rows.
- **site.json:** Array layout, the 3 panels each hub serves, panel ratings, and temperature coefficients
- **weather.csv:** NOAA daily weather for 365 days
- **service_log.csv:** 12 months of past truck rolls, what was found, and what was fixed

3 DELIVERABLES

- **Pipeline code.** 6 Python modules that run with 1 command, plus a README.
- **Site reports.** 2 PDFs (180 and 365 day), each with 6 pages:
 - Summary. Total kWh lost, split into fault and weather, and the top 3 findings.
 - Fault or Weather. Every output drop, its label, and the kWh it cost.
 - Dispatch List. Each panel that needs a truck, the likely cause, and the evidence.
 - Repeat Repairs. Panels that keep getting fixed and the root cause, using the service log.
 - Early Warnings. Panels in slow decline and how fast they are dropping.
 - Data Quality and Method. Usable data percent for all 12 hubs, excluded hubs, and the rules used.
- **Results files.** 5 files, and every number in the reports comes from them.
- **Scorecard.** Faults caught, false alarms, truck rolls avoided, and days of early warning, checked against SolarScope's answer key.
- **Test suite.** Passing tests for all 6 steps, with expected output within 2% of a hand calculation.
- **Process flowsheet.** 1 diagram that matches the code.
- **Final walkthrough.** 1 live run from raw data to a finished PDF with no code changes.